

John Vandermeulen

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Education

Clemson University

B.S. Mechanical Engineering, Minor in Mathematical Sciences

M.S. Mechanical Engineering (Thermal & Fluid Sciences)

Clemson, SC

Dec 2025 | GPA: 3.48

Dec 2026 | GPA: 3.71

Work Experience

Gulfstream Aerospace Corporation

Savannah, GA

Cabin Systems / ACT R&D Co-op (Incoming)

May 2026-Aug 2026

- Incoming member of Advanced Cabin Technologies research and development team
- Group focused on aircraft cabin system integration, testing, and FAA certification support
- Expected to support internal aircraft system design, testing, and engineering analysis

Vendor Project Engineering Co-op

May 2025-Aug 2025

- Developed DO-160 test plans and designed CATIA test fixtures for aircraft components
- Authored 28 technical supplemental reports supporting airworthiness of vendor parts
- Completed over 20 hours of continuous improvement training, earning Yellow Belt certification

Stress Engineer Co-op

Aug 2024-Dec 2024

- Verified structural compliance of 14 custom G700 interior components with FAA Part 25 regulations by coordinating with design engineers and FAA representatives
- Performed hand calculations and FEMAP stress analysis for design validation
- Generated new FEMAP based stress analyses and FAA approved reports using Excel VBA

Baseline Design Engineer Co-op

Jan 2024-Apr 2024

- Completed an intensive, 40-hour advanced CATIA training program
- Designed 37 new workflows including new custom interior asset designs as well as amendments to old engineering using CATIA, AutoCAD, and PLM software ENOVIA SmarTeam
- Worked on long term project to design a new standard interior asset for the G700 program using CATIA, Kaizen walks, and weekly presentations with management

Skills

- CAD & FEA: CATIA, SolidWorks, Onshape, AutoCAD, FEMAP, ANSYS Fluent
- Aerodynamics and Fluid Analysis: XFOIL, Computational Fluid Dynamics (CFD)
- Programming: MATLAB, VBA
- Software and PLM: Excel, Word, PowerPoint, SharePoint, ENOVIA SmarTeam

Specialized Projects

- 2D Viscous CFD Nozzle Solver: Compressible Flow, MacCormack Scheme, Body-Fitted Grid
- Composite Layup Optimization: Laminate Theory, Tsai-Wu Failure, MATLAB Automation
- Turbojet Engine Optimization: Cycle Analysis, Compressor/Turbine Matching
- Aircraft Design Project: Aerodynamics, Performance, and Stability Sizing
- Clemson Rocketry - Propulsion Sub Team: Hybrid Rocket Engine Design and Simulation